# Yang Xiao 肖杨

**FOUNDATION MODEL RESEARCH & ENGINEERING**

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Website Scholar GitHub

**8** first-author papers | **5** accepted at NeurIPS · AAAI · ACL · NAACL | **1,003** citations | **2×** ACL best awards

## PROFILE

I build foundation models through large-scale continued pre-training (CPT), reasoning post-training, and agentic post-training. At XHS, I lead end-to-end data, training, and evaluation for large-scale user modeling and code agents, scaling CPT to 100K users and 200B tokens.

## EDUCATION

**The Hong Kong Polytechnic University**, Hong Kong (2023–July 2027) – Ph.D. in Computing; advisors: Prof. Wenjie Li and Prof. Pengfei Liu; Presidential PhD Fellowship (top 1%).

**Fudan University**, Shanghai (2017–2022) – B.E. in Software Engineering; GPA rank: top 1%; National Scholarship (top 1%).

## EXPERIENCE

### Xiaohongshu (REDnote)

Mar 2026 – Present

ACE Talent Program, Foundation Model Team

#### Large-Scale User-Behavior Modeling with LLMs

Project lead

**(1)** Task synthesis: Designed 100+ task templates spanning domains, task types, prediction targets, ground-truth sources, answer formats, and scoring rubrics, scaling data generation to **100K users, 90 domains, 900K tasks, and 200B tokens**. **(2)** Task filtering: Screened synthesized tasks against task-specific constraints, ground truth, and scoring rubrics, removing invalid or ambiguous instances before rollout. **(3)** Difficulty-aware rollout: Ran four student-model rollouts per task and removed tasks that were either too easy or too hard; ran four teacher-model rollouts for each remaining task and retained the highest-scoring trajectory. **(4)** Trajectory filtering: Kept only complete, correctly formatted, and verifiable trajectories; filtered bad reasoning patterns such as repeated segments, n-gram repetition, and periodic loops, using an LLM judge for borderline cases. **(5)** Evaluation system: Built frozen in-domain and out-of-domain suites spanning IFBench, AIME, GPQA, MMLU-Pro, and in-house benchmarks; monitored accuracy, response-length distributions, format validity, and repetition to ensure performance gains without introducing degenerate behaviors. **(6)** Insight & results: Observed simultaneous in-domain and out-of-domain gains, showing that the data recipe improves transferable capabilities rather than overfitting to in-domain tasks. **(7)** Data integration: Owned final data consolidation into the base-model training mixture and supported the resulting model release.

#### Code-Agent Model Optimization

**(1)** Code-agent infrastructure: Co-built sandbox infrastructure and the end-to-end Sandbox × Harbor × Task × Harness × LLM pipeline, supporting automated high-concurrency rollouts and model evaluation. **(2)** Task synthesis: Built a compositional generator that independently samples nine axes—domain, skill type, primitive skills, persona, language, task complexity, command complexity, fixture type, and verifier type—to create **20K+ executable environments** in Harbor task format. **(3)** Trajectory quality: Rolled out and curated **5K+ verifier-checked trajectories**; removed low-quality or degenerate runs, including incomplete executions, failed verification, and repetitive thinking patterns. **(4)** Training result: Raised Qwen3.5-35B-A3B on Terminal-Bench 2.1 from **44 to 62**, and raised Qwen3.5-397B-A17B from **52.5 to 70** on the same benchmark.

#### PILOT in the Loop: Live Self-Improvement for Long-Horizon Agents

First author · Preprint

**(1)** Harness architecture: Built a separate supervisor–worker architecture with a two-way live channel: workers notify, ask, and return results; supervisors steer or abort active runs, then distil procedures and failure modes into persistent skills and memory. **(2)** One-shot setting: Across two frozen backbones and three long-horizon benchmarks, PILOT ranks first in 5 of 6 configurations and leads Terminal-Bench 2.0 by up to **9.8 points**. **(3)** Self-improvement setting: Improves pass rate by **12.4–14.6 points**, reduces output tokens by 42.9–47.4%, and increases successful runs per 1M output tokens by **110.3–134.0%**.

## EXPERIENCE

**Shanghai Innovation Institute** Sep 2025 – Feb 2026
Algorithm Intern

### Code-Agent Model Optimization

First author · Preprint

**(1)** Real-world tasks: Curated authentic vibe-coding and scientific-research requests from professional developers and researchers, preserving long-horizon planning, tool use, and collaboration. **(2)** PR-based synthesis: Built a GitHub PR query-synthesis pipeline over 100 repositories with 10K+ stars each; diversified development domains and applied code-change complexity filters and expert review. **(3)** Trajectory curation: Collected **3.3M high-quality long-horizon trajectory tokens** in the SII CLI environment, preserving complete reasoning, tool calls, environment feedback, error recovery, and human–agent collaboration (42.4K-token average; 152K maximum).
**(4)** Cross-scale post-training: Fine-tuned GLM-4.5-Air (106B) and GLM-4.5 (355B) with a controlled slime-based SFT setup. **(5)** Evaluation system: Evaluated AgencyBench, tau2-bench, EvalPlus, DS-1000, and SciCode under both CLI-enabled and tool-free settings to separate tool-use gains from intrinsic capability gains. **(6)** Results & insight: Reached **73.5** on AgencyBench and 57.2 across generalization benchmarks; improved the 106B and 355B base models by 17.3 and 28.4 points, and outperformed a 25M-token SFT baseline by **53.7%**, showing that carefully curated long-horizon data transfers across model scales and domains. Code

## FIRST AUTHOR

**PILOT in the Loop: Live Self-Improvement for Long-Horizon Agents** Preprint · 2026
Yang Xiao et al.

A supervisor–worker harness couples live steering with live self-evolution for long-horizon agents.

**LIMI: Less is More for Agency** Preprint · 2025
Yang Xiao et al.

Agent post-training from high-quality long-horizon trajectories. Code

**LIMOPro: Reasoning Refinement for Efficient and Effective Test-time Scaling** NeurIPS · 2025
Yang Xiao, Jiashuo Wang, Ruifeng Yuan, Chunpu Xu, Kaishuai Xu, Wenjie Li, Pengfei Liu

Improves reasoning accuracy by up to 6.6% while using up to 41% fewer tokens. Code

**SCALE: Selective Resource Allocation for Overcoming Performance Bottlenecks in Mathematical Test-time Scaling** AAAI · 2026
Yang Xiao, Chunpu Xu, Ruifeng Yuan, Jiashuo Wang, Wenjie Li, Pengfei Liu

Allocates inference compute according to sub-problem difficulty, improving Qwen2.5-32B on AIME24 by 35.4 points. Code

**Towards Dynamic Theory of Mind: Evaluating LLM Adaptation to Temporal Evolution of Human States** ACL · 2025
Yang Xiao et al.

Long paper at the ACL main conference.

**DataLab: A Platform for Data Analysis and Intervention** ACL · 2022
Yang Xiao et al.

Outstanding Demo Paper Award.

**Are All the Datasets in Benchmark Necessary? A Pilot Study of Dataset Evaluation for Text Classification** NAACL · 2022
Yang Xiao et al.

Studies benchmark composition and individual dataset contribution.

**How Far Are LLMs from Believable AI? A Benchmark for Evaluating the Believability of Human Behavior Simulation** Preprint
Yang Xiao et al.

Benchmarks the believability of simulated human behavior.

## SELECTED CO-AUTHOR

**LIMO: Less is More for Reasoning** COLM · 2025
Yixin Ye, Zhen Huang, Yang Xiao et al. · Third author

527 citations and 1.1k GitHub stars. Code

**ExplainaBoard: An Explainable Leaderboard for NLP** ACL · 2021
Co-author

Best Demo Award.

## HONORS

**PolyU Presidential PhD Fellowship** · top 1% · 2023–2027

**Outstanding Demo Paper Award**, ACL · DataLab · 2022

**Best Demo Award**, ACL · ExplainaBoard · 2021